The Paradox of Authorial Identity: Measuring Style Decay and Semantic Retention Under Iterative LLM Paraphrasing

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ABSTRACT

A machine paraphrase keeps *what* a text says while rewriting *how* it says it. We ask a Ship-of-Theseus question for authorship: after three rounds of LLM paraphrasing, does a document still carry its author's identity? On 700 human-written documents from seven domains of the Ship of Theseus corpus, paraphrased by seven LLM settings and a rule-based baseline, we find that wording is replaced in a single pass while meaning is largely kept. Comparing ten lexical, syntactic and semantic feature families on a common scale (retention relative to an unrelated document), style is replaced long before meaning: the syntactic skeleton goes as fast as the lexical skin, and each paraphraser overwrites the author's style with its own.

1 PROBLEM DESCRIPTION

A machine paraphrase preserves *what* a text says while rewriting *how* it says it. Applied repeatedly, this poses the Ship of Theseus question for authorship: after a document has been paraphrased three times by a large language model, whose text is it? Authorship attribution, machine-generated text (MGT) detection, plagiarism analysis, and forensic provenance all assume a stable stylistic signal survives in a document; paraphrasing cheaply defeats that assumption.

We therefore measure identity decay *quantitatively* rather than as a single pass/fail outcome, tracking generation by generation which linguistic properties are destroyed first and which persist. Three questions organize the work: **(RQ1)** Which stylometric features decay before semantic content does? **(RQ2)** At which generation does authorship attribution become unreliable? **(RQ3)** Does the paraphraser leave a recoverable fingerprint, so that authorial identity is not merely erased but *replaced*? Our contribution is a decay curve that separates human-author signal loss from model-fingerprint gain. Code, data pipeline and results: https://github.com/gajawp/ShipOfThesus.

2 BACKGROUND

Iterative paraphrasing and authorship. In the work done by Tripto et al. [11] at ACL 2024, the authors ask whether a text keeps its authorship after repeated paraphrasing. They apply four paraphrasers (ChatGPT, PaLM2, Pegasus, DIPPER) three times over seven datasets and re-run attribution and machine-generated-text (MGT) detectors at each generation. Attribution accuracy degrades with every round, but only aggregate accuracy is reported, so *which* features decay is never isolated—the gap our work addresses.

Paraphrase as an evasion attack. In the work done by Krishna et al. [4] at NeurIPS 2023, the concern is the robustness of AI-text detectors. They train DIPPER, an 11B-parameter discourse-level paraphraser with controllable lexical and order diversity, and propose a retrieval defense. One DIPPER pass drives DetectGPT and GPTZero toward chance, but the defense assumes the generating API logs its outputs.

Zero-shot MGT detection. In the work done by Mitchell et al. [6] at ICML 2023, the goal is to detect generated text without labeled data. DetectGPT compares a text's log-likelihood with that of small perturbations of it, exploiting negative curvature of the model's probability surface. It beats prior zero-shot baselines but needs white-box likelihoods and collapses under paraphrasing.

Attribution across neural generators. In the work done by Uchendu et al. [12] at EMNLP 2020, authorship attribution is extended from human writers to neural text generators. They benchmark stylometric, deep-learning and transformer classifiers on three attribution tasks over nineteen generators. Generators are separable, but only single-pass text is tested, leaving open whether signatures survive rewriting.

Content-independent author representations. In the work done by Rivera-Soto et al. [10] at EMNLP 2021, the question is whether one authorship representation transfers across domains. They train a Transformer encoder contrastively on Reddit, Amazon and fan-fiction so that same-author documents embed together. The embeddings transfer well but remain sensitive to topic overlap, a style–content confound that our fixed-source design removes.

Measuring semantic retention. In the work done by Zhang et al. [14] at ICLR 2020, the problem is evaluating generated text beyond n-gram overlap. BERTScore greedily matches candidate and reference tokens by contextual-embedding similarity. It tracks human adequacy judgments far better than BLEU [7], but it is blind to style, so we pair it with explicit stylometry.

Distributional gap between neural and human text. In the work done by Pillutla et al. [8] at NeurIPS 2021, the problem is measuring how far a model's text distribution is from human text. MAUVE compares the two distributions in an embedding space using divergence frontiers. It correlates well with human quality judgments, but it scores whole corpora, not individual documents, so it cannot trace one text's identity across generations.

Watermarking LLM outputs. In the work done by Kirchenbauer et al. [3] at ICML 2023, the goal is to make LLM text verifiably detectable. They bias sampling toward a pseudo-random "green" token list and detect it with a statistical test. The watermark survives light edits but weakens under paraphrasing, which motivates asking whether the paraphraser leaves a fingerprint of its own (RQ3).

Detection and human judgment. In the work done by Ippolito et al. [2] at ACL 2020, the authors study how decoding strategy affects human and automatic detection of generated text. They benchmark three sampling methods with expert human raters and fine-tuned BERT classifiers. Sampling that fools humans makes text easier for classifiers to detect, so both are needed; yet both rely on surface cues that a paraphrase rewrites.

Visual statistical detection. In the work done by Gehrmann et al. [1] at ACL 2019 (System Demonstrations), the goal is to help humans spot generated text. GLTR colours each token by its rank under a language model, exposing unusually predictable text. It raised human detection rates from 54% to 72%, but its token-level signal is exactly what a paraphrase replaces.

3 DATASET

3.1 The Ship of Theseus Corpus

We use the paraphrased corpus released by Tripto et al. [11], organised along three axes:

- **Seven datasets (domains):** XSUM (BBC news), Yelp (reviews), CMV (Reddit ChangeMyView opinions), ELI5 (Reddit explanatory answers), SciGen (scientific text), TLDR (Reddit summaries) and WP (Reddit WritingPrompts stories).
- **Seven sources** that wrote the original T0: Human and six LLMs (OpenAI, LLaMA, PaLM, Tsinghua, BigScience, Eleuther-AI) prompted with the first 30 tokens of the human text, so human and LLM texts from one prompt share a document `key`.
- **Seven paraphraser settings** of four models, each applied three times (T0→T1→T2→T3): ChatGPT (GPT-3.5-turbo), PaLM2, Pegasus [13] (*full*: every sentence; *slight*: 25% of sentences) and DIPPER [4] (default, *low* and *high* diversity).

Each dataset has two files (14 in total). The **paraphrased** file (≈420k rows) holds every original with its 21 paraphrases, with columns `source`, `key`, `text` and `version_name` (`original`, `chatgpt`, `chatgpt_chatgpt`, ...). The **train** file (≈19.5k rows) holds un-paraphrased texts from the same sources; we verified that it shares no key with the paraphrased file. The paraphrased files serve RQ1, RQ3 and the RQ2 test set; the train files train the RQ2 attribution models. As a controlled reference we add a rule-based **WordNet** baseline that replaces 25/38/50% of content words with same-POS synonyms at T1/T2/T3.

Table 1: Human T0 sample (100 documents per domain).

Domain	Avg. words	Avg. sents	Vocab
XSUM (news)	300.0	15.1	5,952
Yelp (review)	131.5	9.4	2,677
CMV (opinion)	401.8	23.9	4,899
ELI5 (QA)	242.2	12.5	4,126
SciGen (science)	163.0	7.0	3,181
TLDR (summary)	84.7	4.8	2,458
WP (story)	418.3	33.7	6,280
All (700 docs)	248.8	15.2	15,888

3.2 Preprocessing

All 14 files pass through one logged pipeline (src/data/preprocess.py): **(1) Cleaning**: HTML entities decoded, Unicode NFKC-normalised, curly quotes and dashes mapped to ASCII, whitespace collapsed. Sources format text differently (DIPPER outputs begin with a space; some models emit curly quotes), so without this step punctuation features would measure formatting rather than style; ≈44% of rows changed. We deliberately do *not* lower-case, remove stop-words, stem or truncate, since that would destroy the stylistic signal under study. **(2) Chain alignment**: one row per (source, key) with T0 and all 21 paraphrases, keeping only documents complete for all seven settings so every paraphraser is compared on the same texts (retention 90–97%; SciGen 58% and WP 69% because many PaLM chains are missing). **(3) Length filter**: T0 under 30 words removed (≈2%). **(4) Sanity checks**: T1 outputs sharing <10% of their words with T0 are flagged. This exposed a corpus artifact: in five of seven domains 54–95% of **DIPPER (high)** outputs are unrelated to their source (e.g. narrative fiction in place of a BBC article), so we exclude that setting from interpretation. **(5) Leakage-free splits**: train files split 85/15 by key, so a human text and the LLM texts from its prompt never straddle the split.

The result is 16,386 aligned chains (2,269 Human) and 19,200 training texts. Because RQ1 concerns the decay of a human author's identity, the analyses below use a fixed sample of **100 Human documents per domain** (700 in total, seed 42; Table 1).

4 EXPERIMENTATION AND INITIAL RESULTS

4.1 Setup

Each Tk is compared with its own T0 (not with T$k-1$), so scores express cumulative divergence from the human original, averaged over the 700 documents. **Wording kept:** BLEU-4 [7] (smoothed) and ROUGE-L F1 [5]. **Meaning kept:** BERTScore F1 [14] (distilbert-base-uncased) and SBERT cosine [9] (all-MiniLM-L6-v2). Raw BERTScore gives even unrelated text ≈0.67, so we rescale it with the official baseline, $(F - 0.666)/(1 - 0.666)$, mapping unrelated text to ≈0. A faithful paraphrase scores low on BLEU and ROUGE-L, so we read them only as measures of surface replacement.

Table 2: Similarity to T0 for ChatGPT chains (700 Human docs). BERTSc. is rescaled so unrelated text ≈ 0.

Gen.	BLEU-4	ROUGE-L	BERTSc.	SBERT
T1	0.192	0.506	0.653	0.900
T2	0.164	0.461	0.615	0.884
T3	0.137	0.425	0.592	0.875
Drop T0→T3	*86%*	*58%*	*41%*	*12%*

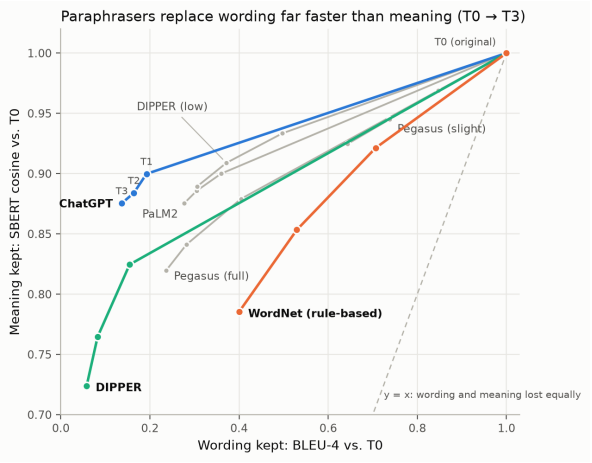


Figure 1: Path T0→T3 per paraphraser: wording kept (BLEU-4) vs. meaning kept (SBERT). All paths stay far above $y = x$. DIPPER (high) omitted (unrelated outputs).

4.2 Wording Is Replaced, Meaning Stays

For ChatGPT (Table 2), BLEU-4 loses 86% of its value while SBERT loses only 12%, and 94% of the BLEU drop already happens at T1; later rounds change little. Figure 1 shows that **every paraphraser discards wording faster than meaning**, but at different rates. ChatGPT keeps the same meaning as PaLM2 at T3 (SBERT 0.875 vs. 0.876) with half its word overlap (BLEU 0.14 vs. 0.28): it rewrites most aggressively at no semantic cost. Default DIPPER loses the most of both (BLEU 0.06, SBERT 0.72). The rule-based WordNet baseline moves the opposite way to the LLMs, keeping the most wording (BLEU 0.40) but losing more meaning (SBERT 0.79): synonym swaps keep the skeleton and break the sense, so simulated paraphrasing is no substitute for real LLM chains. DIPPER (high) scores SBERT 0.06, the level of unrelated text, confirming the preprocessing check.

Domain matters (Table 3). Science, summaries and news keep the most meaning: terms, entities and numbers must be copied, which anchors the content. Stories and opinions keep the least, since they leave room to restructure narrative and rhetoric. The effect depends on the paraphraser: PaLM2 copies heavily in science (BLEU 0.50) but rewrites opinion almost fully (0.11), while DIPPER-low is nearly flat across domains.

Table 3: SBERT similarity to T0 at T3 by domain (100 docs each), one setting per model.

Domain	ChatGPT	PaLM2	DIPPER-low	Peg.-full
Science	0.947	0.969	0.925	0.857
Summary	0.924	0.944	0.906	0.813
News	0.888	0.912	0.870	0.787
Review	0.854	0.859	0.887	0.831
QA	0.850	0.828	0.867	0.827
Opinion	0.848	0.808	0.906	0.822
Story	0.817	0.810	0.863	0.801

4.3 RQ1: Which Planks Are Replaced First?

Corpus averages hide change (a feature can keep its mean while every document moves), so we compare each document with its own T0 on ten feature families, computed from one spaCy parse. *Lexical (skin):* word choice (Jaccard), 198-word function-word profile, 12-mark punctuation profile, lexical richness (MATTR, word length). *Syntactic (skeleton):* sentence shape (mean, s.d. of length), POS distribution, dependency-label distribution, tree depth (mean/max depth, dependency distance), Flesch–Kincaid grade. *Meaning (hull):* SBERT. Distributions are compared with Jensen–Shannon distance, scalars with the RMS difference of z-scores.

One scale. Each distance is divided by the distance between the T0 and five unrelated T0s of the same domain, giving retention

$$R = 1 - d(\mathrm{T}k, \mathrm{T0}) \,/\, d(\text{unrelated}, \mathrm{T0}).$$

$R = 1$ means unchanged, $R = 0$ as different as another author's document, and $R < 0$ pushed beyond it. We report 95% bootstrap CIs and paired one-sided Wilcoxon tests (style distance > meaning distance) with Holm correction.

Style goes, meaning stays (Figures 3 and 2). For ChatGPT every style family keeps ≤0.34 at T3 (most ≈0.2) while meaning keeps 0.85 (95% CI 0.85–0.86). Meaning is the most retained family for every LLM paraphraser, and in 61 of 63 paraphraser × family tests at T1 the style distance is significantly larger than the meaning distance (Holm $p < 0.05$); both exceptions are WordNet (function words, sentence shape), whose word swaps leave those features in place. **Gone in the first pass.** For ChatGPT and PaLM2, 82–92% of each style family's three-round loss happens at T1. **The skeleton goes with the skin.** Syntax is not more durable than vocabulary: mean lexical vs. syntactic retention is 0.15 vs. 0.14 for ChatGPT and 0.33 vs. 0.30 for PaLM2. Pegasus-full restructures most (syntactic −0.53; tree depth −2.27).

Overwritten, not just erased. Negative retention marks a consistent shift in one direction (Table 4). ChatGPT formalises (reading grade 9.8→12.4, $d_z = 1.05$; word length $d_z = 1.50$), DIPPER does the reverse (shorter words, more determiners), and Pegasus splits sentences and flattens parse trees ($d_z = -1.36$). Because the direction identifies the paraphraser, this is early evidence for RQ3: authorial identity is not merely erased but replaced.

Preliminary answer to RQ1. Under LLM paraphrasing, style is replaced long before meaning: after one pass of ChatGPT, PaLM2 or DIPPER the style families have moved 55–90% of the way to an

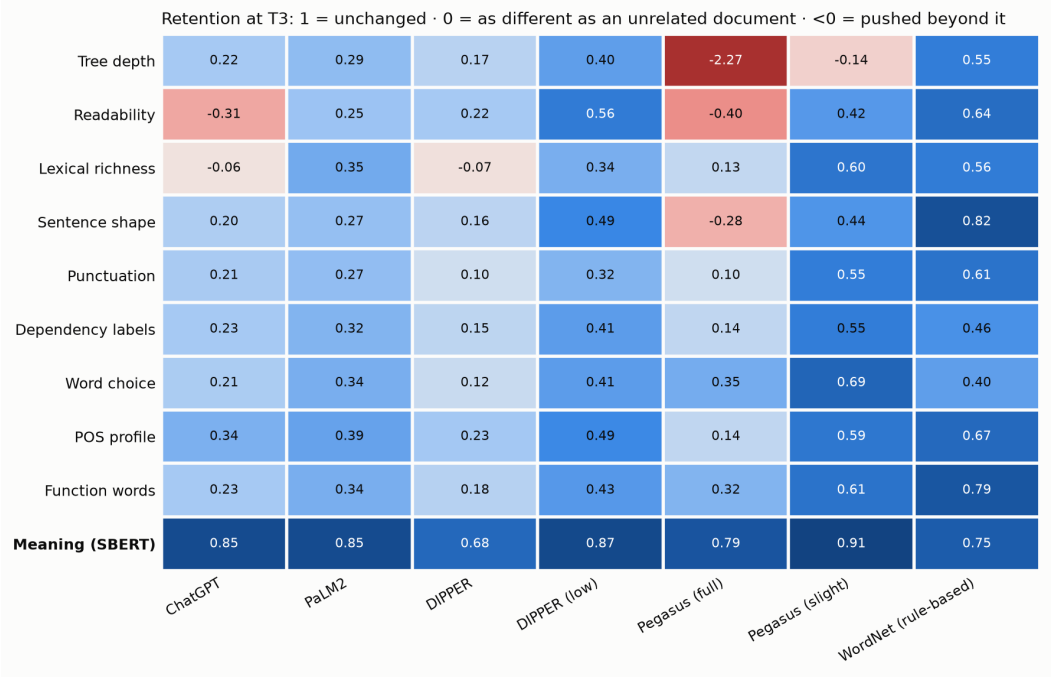


Figure 2: Retention at T3 for every feature family and paraphraser (700 docs; DIPPER-high excluded). Blue = share of the original signal kept, red = pushed beyond an unrelated document. Meaning (bottom row) is the most retained family for every LLM paraphraser.

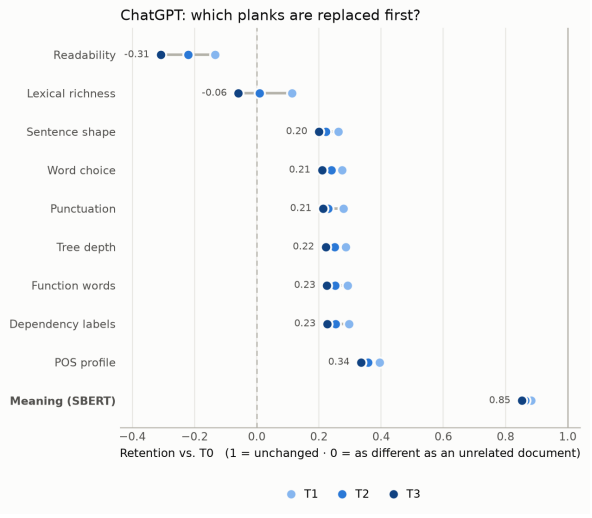


Figure 3: ChatGPT: retention of each family at T1–T3 (700 docs), first-replaced on top. Style is mostly gone after one pass; readability and lexical richness go below 0.

unrelated author's style while 79–88% of the meaning is retained, and the syntactic skeleton is replaced as fast as the lexical skin.

Table 4: Mean T0→T3 change. ↑ /↓: Holm-significant with $|d_z| \geq 0.5$.

Feature	T0	ChatGPT	DIPPER	Peg.-full
Reading grade	9.8	12.4↑	9.2	6.6↓
Word length	4.68	5.21↑	4.42↓	4.47↓
MATTR	0.81	0.83↑	0.77↓	0.78↓
Sentence length	18.3	18.9	19.0	12.7↓
Tree depth	5.2	5.7	5.1	4.0↓
Pronouns	.079	.064↓	.087	.081
Determiners	.083	.087	.105↑	.094

5 PLAN FOR NEXT STEPS

Limitations to address. (i) Only Human-written T0 has been analysed; the six LLM sources are preprocessed and will be compared next, including whether a model changes its own text less (OpenAI T0 under ChatGPT, PaLM T0 under PaLM2). (ii) Paraphrases are shorter than their sources and short texts give noisier profiles, so we will add a length-matched baseline. (iii) SBERT reads only 256 tokens, which truncates 43% of documents; a chunked encoding is implemented, and a first check raises SBERT at T3 by 0.01–0.05 in long-text domains (ChatGPT on stories: 0.817→0.861), so final results will use it.

Weeks 9–11: Authorship attribution (RQ2). Train a stylometric SVM and a fine-tuned RoBERTa to identify the seven sources on the train files, evaluate them on T0–T3, and locate the "point of no return" per paraphraser.

Weeks 10–12: Paraphraser fingerprinting (RQ3). Train a paraphraser classifier on T1–T3 (excluding DIPPER-high) with a key-level split, and visualise identity trajectories with t-SNE and quantitative cluster-separation measures.

Weeks 13–14: Synthesis. A Streamlit decay dashboard and the final paper.

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